



Agentic AI Foundation

AGNTCon + MCPCon China

Where the **agentic stack** is being built.

Mission-Critical Agentic OS Runtime

CHAO LIU

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Long-Task Performance

COMPLETION QUALITY · PARAMETERS REQUIRE CALIBRATION

$$CQ_{100} \approx (1 - D_{100}) \times A(N)$$

QUALITY DRIVERS

1-D

Low drift →
high completion

Goal continuity

N

Agent Check

More → better

A(N)

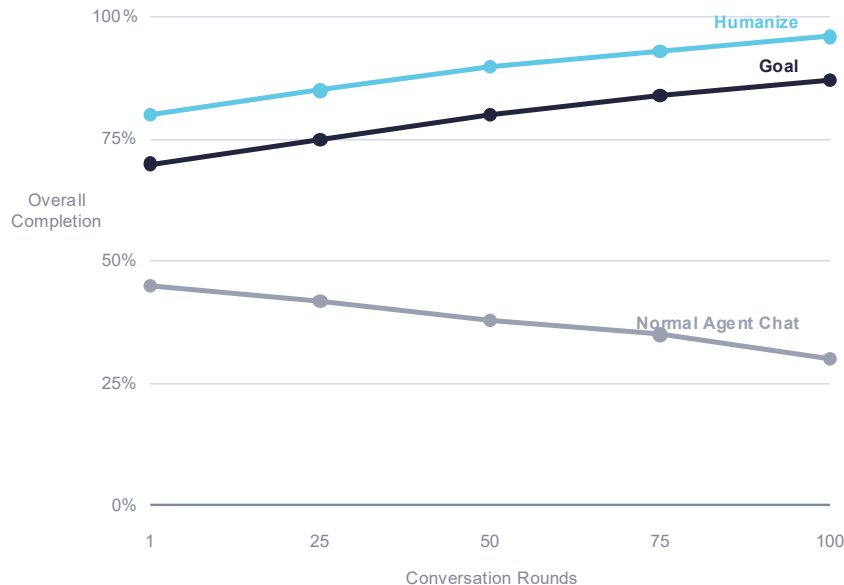
Check
precision

Quality ↑

Goal Completion ≠ High Quality

Humanize: Builder × Reviewer cross-check

Conversation Rounds × Completion Quality



Conceptual illustration · parameters TBD

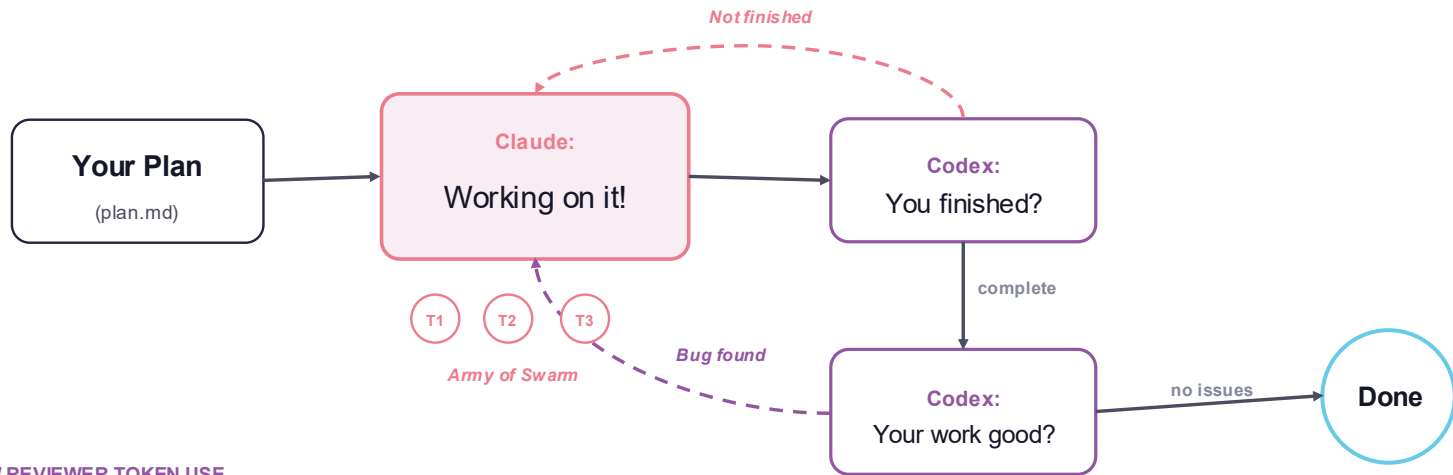
Humanize 1.0



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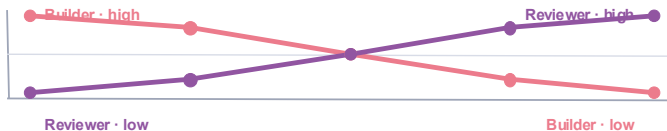
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RLCR · RALPH-LOOP WITH CODEX REVIEW



BUILDER / REVIEWER TOKEN USE

Relative trend across one RLCR run · not an absolute benchmark



Where the **agentic stack** is being built.

Humanize 2.0



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HUMANIZE 2.0 · RUNTIME IN ACTION

One flow. Many agents. One trace.

Humanize 2.0 turns a directory of Python into the runtime: it selects the agents, the order, and when to stop.

Python ≥ 3.12 · drives your CLI · no API key

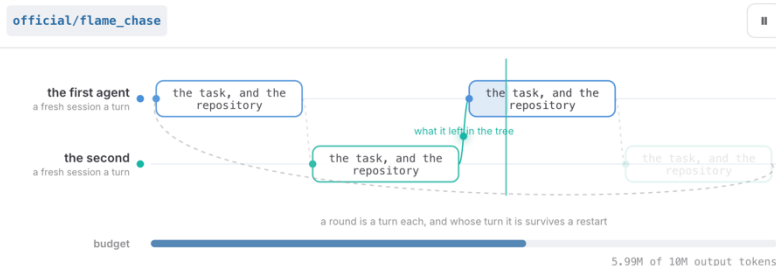
10 CLIs

container / ssh

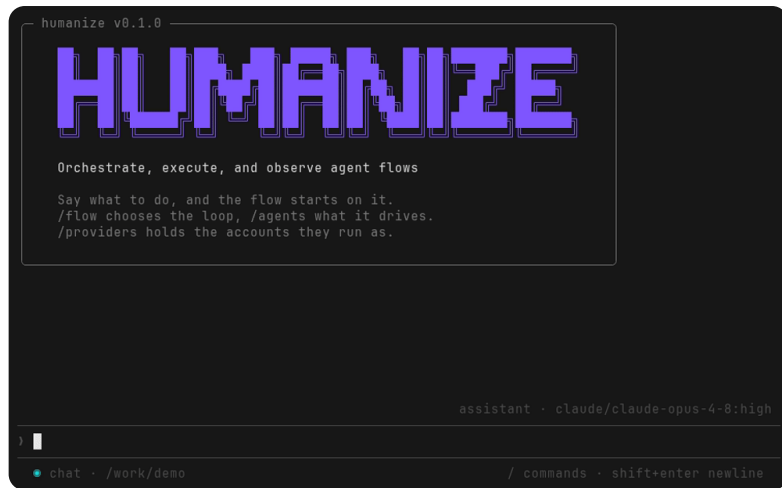
Perfetto trace

RUN A FLOW

hmz exec -f official/flame_chase



TUI EXECUTION



KEY WORKFLOWS

4 KEY PATTERNS

official/humanize1.ricr
builder + reviewer · persist until clean

official/r1ar
actor + fresh reviewer · review ends run

official/flame_chase
2 agents · alternate on one working tree

official/parallel_flame_chase
7 agents · 3 isolated lanes

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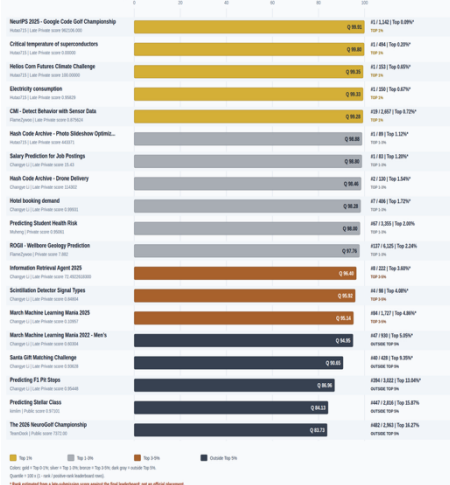
Humanize: Practice Results

KDA (Kernel Design Agents)

Best tracked result by completed competition

Foraground agents | gold when better at Top 25 | 25% / 1%

28 September 2024 00:00 to 12 October 2024 00:00 UTC+8



1.17–1.69× SOTA

TOP 3 / ALL TRACKS
SOLExecBench L1 · #1
Repeatable kernel loop
Profile → verify

Beni / Robot Productization



4K · AUTO-FOLLOW · HDR

Tracks moving subjects.
HDR color + auto-edited highlights.
10-inch obstacle jump.

Rein: Edge Agent Harness

REIN · AGENT HARNESS ARCHITECTURE

Endpoint front door. Preset-first composition.

ARCHITECTURE



WHAT REIN ADDS

A small terminal Agent for the edge.

FRONT DOOR

line-oriented REPL + one-shot ask

PRESET-FIRST

persona + tools + context per agent

EDGE-READY

pure JS · Node ≥ 24 · x86-64 / ARM64 / RISC-V64

Rein owns the front door · DeepSeek Harness owns execution.

Where the **agentic stack** is being built.

Agentic OS Runtime

AGENTIC OS RUNTIME · COMPOSED FROM BOTTOM TO TOP

MONITOR / AUDIT

ASYNC CONTROL

HUMANIZE 2.0 · DYNAMIC WORKFLOW

dynamic task planning · orchestration · resume

RUNTIME LAYER

AGENTIC OS RUNTIME

Rein

Agent Preset

Permission

OTHER MIDDLEWARE

DDS

ROS 2

microXAgent

LINUX OS

X86-64

ARM64

RISC-V 64

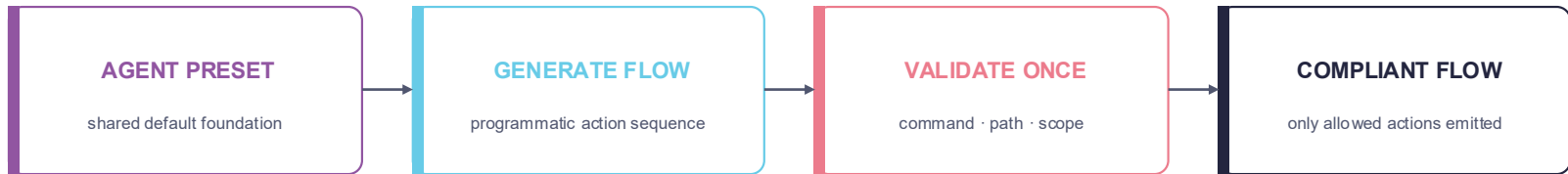
Where the **agentic stack** is being built.

Permission-Aware Flow

AGENTIC OS RUNTIME · FLOW GENERATION

We do not check each command at execution time.
We generate a permission-compliant flow up front.

GENERATE ONCE / EXECUTE MANY



EXECUTE THE VALIDATED FLOW



The compliant flow becomes the foundation for every Rein Agent Preset default.

Security Audit

ACTION PATH / EVENT STREAM

AUDIT UPGRADE / REPLAYABLE

Every agent action leaves a trace we can audit.



PATH UPGRADE / HUMAN GATE

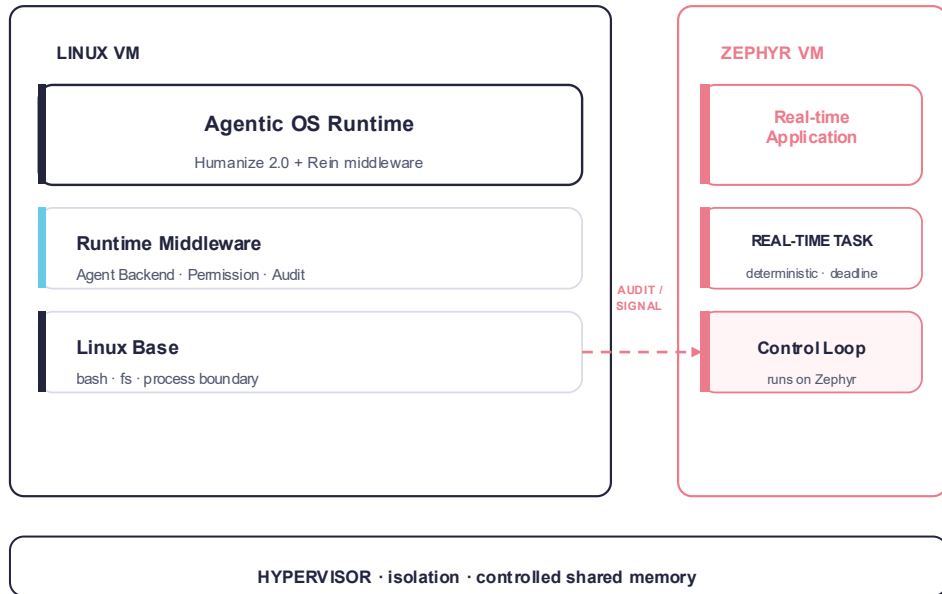


AUDIT PATHS · <project>/rein/{sessions,traces} · ~/.rein/effect-ledger.json

Agent handles L0–L1; human approval gates L2.

Demo : Linux + Zephyr

SYSTEM PATH



CURRENT DEMO / ESTIMATED EFFECT

vs. ordinary agent harness

TASK QUALITY

+15–25%

completion confidence



OPTIMIZATION LOOP

+20–30%

iteration throughput



TOKEN SAVINGS

–20–35%

context re-entry



Illustrative estimate · validate on Linux + Zephyr

Next Plan

01

Permission Matrix

Bash · FS · device

02

Audit Replay

Trace · hash · resume

03

Linux + Zephyr

VM · RTOS · hypervisor

04

Benchmark

Edge OS · Agent Harness



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OPEN SOURCE

REIN <https://github.com/processmission/rein>

HUMANIZE <https://github.com/PolyArch/humanize>

HUMANIZE 2 <https://github.com/humanfia/humanize2>